ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Technical Programming paper 2 · practice e.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Programs are not marked strictly on syntax: give marks for correct logic, correct variables and correct output. Accept other variable names, prompts and correct ways of solving the problem.

QUESTION 1

1.1 Project created and named ✓ (1)

1.2 Suitable sprite ✓ (1)

1.3 Suitable backdrop ✓ (1)

1.4 Variables amount, days, units, perDay created ✓✓✓✓ and set to 0 ✓✓✓✓ (8)

1.5 ask block ✓ with a clear question ✓; set amount to answer ✓ (3)

1.6 ask block ✓ with a clear question ✓; set days to answer ✓ (3)

1.7 units = amount ÷ 2.5 ✓✓; perDay = units ÷ days ✓✓ (4)

1.8 if perDay < 10 ✓ "Careful…" ✓; else if perDay > 20 (or ≤ 20) ✓ correct branches ✓; "Good" ✓ and "Plenty" ✓ (6)

1.9 say units ✓, units per day ✓ and advice ✓ (3)

1.10 200 units, 6,67 per day – Careful; 400 units, 13,33 per day – Good; 300 units, 30 per day – Plenty ✓✓ (2)

```
when green flag clicked
set [amount] to 0
set [days] to 0
set [units] to 0
set [perDay] to 0
ask [How much money will you spend (R)?] and wait
set [amount] to (answer)
ask [For how many days must it last?] and wait
set [days] to (answer)
set [units] to ((amount) / 2.5)
set [perDay] to ((units) / (days))
say (join (join [Units: ] (units)) (join [  Per day: ] (perDay))) for 3 seconds
if < (perDay) < 10 > then
    say [Careful, save electricity] for 2 seconds
else
    if < (perDay) > 20 > then
        say [Plenty] for 2 seconds
    else
        say [Good] for 2 seconds
```

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QUESTION 2

2.1 File created and named correctly ✓ (1)

2.2 #include <iostream> ✓ (and <string>); using namespace std; ✓ (2)

2.3 int main() ✓; celsius, fahrenheit, kelvin as double ✓✓; category as string ✓ (4)

2.4 Prompt ✓ and cin ✓ (2)

2.5 fahrenheit = celsius * 9 / 5 + 32 ✓✓✓ (using 9.0 / 5 or double to avoid integer division); kelvin = celsius + 273.15 ✓✓ (5)

2.6 if (celsius < 0) ✓ "Freezing" ✓; else if (celsius <= 25) ✓✓ "Mild" ✓; else ✓ "Hot" ✓; correct structure ✓ (8)

2.7 Fahrenheit ✓, kelvin ✓ and category ✓ displayed with labels (3)

```
// YourName - Question 2
#include <iostream>
#include <string>
using namespace std;

int main() {
    double celsius = 0, fahrenheit = 0, kelvin = 0;
    string category = "";

    cout << "Enter the temperature in degrees Celsius: ";
    cin >> celsius;

    fahrenheit = celsius * 9.0 / 5 + 32;
    kelvin = celsius + 273.15;

    if (celsius < 0) {
        category = "Freezing";
    } else if (celsius <= 25) {
        category = "Mild";
    } else {
        category = "Hot";
    }

    cout << "Fahrenheit: " << fahrenheit << endl;
    cout << "Kelvin: " << kelvin << endl;
    cout << "Category: " << category << endl;
    return 0;
}
```

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QUESTION 3

3.1 File created and named correctly ✓ (1)

3.2 #include <iostream> ✓; using namespace std; ✓ (2)

3.3 int main() ✓; number of runners ✓; total and average (double) ✓; fastest, slowest and counter ✓ (4)

3.4 Prompt ✓ and cin ✓ in a loop ✓ that repeats while n < 1 or n > 20 ✓ (4)

3.5 double array ✓ of size 20 ✓ declared before use ✓ (3)

3.6 for loop ✓✓; prompt with the runner number ✓; cin into times[i] ✓; inner loop / check ✓ while the time is 0 or less ✓ with an error message ✓; correct index use ✓ (8)

3.7 Total set to 0 and each time added ✓✓; average = total / n ✓✓ (4)

3.8 fastest and slowest start at times[0] ✓✓; loop ✓; if times[i] < fastest ✓ then fastest = times[i] ✓; if times[i] > slowest ✓ then slowest = times[i] ✓; correct results ✓ (8)

3.9 Counter for times < 30 ✓✓; loop that displays each time greater than the average ✓✓ (4)

3.10 Average ✓, fastest ✓, slowest ✓ and number under 30 ✓ displayed with labels ✓ (5)

```
// YourName - Question 3
#include <iostream>
using namespace std;

int main() {
    int n = 0, under30 = 0;
    double times[20], total = 0, average = 0, fastest = 0, slowest = 0;

    do {
        cout << "How many runners (1-20)? ";
        cin >> n;
    } while (n < 1 || n > 20);

    for (int i = 0; i < n; i++) {
        cout << "Time for runner " << i + 1 << " (min): ";
        cin >> times[i];
        while (times[i] <= 0) {
            cout << "Time must be more than 0." << endl;
            cout << "Time for runner " << i + 1 << " (min): ";
            cin >> times[i];
        }
    }

    fastest = times[0];
    slowest = times[0];
    for (int i = 0; i < n; i++) {
        total += times[i];
        if (times[i] < fastest) fastest = times[i];
        if (times[i] > slowest) slowest = times[i];
        if (times[i] < 30) under30++;
    }
    average = total / n;

    cout << "Average time: " << average << " min" << endl;
    cout << "Fastest: " << fastest << " min   Slowest: " << slowest << " min" << endl;
    cout << "Under 30 minutes: " << under30 << " runners" << endl;
    cout << "Slower than average:";
    for (int i = 0; i < n; i++) {
        if (times[i] > average) cout << " " << times[i];
    }
    cout << endl;
    return 0;
}
```

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TOTAL: 100

END OF MARKING GUIDELINE